

LAB SESSION 8

Lab Name: Verification, Validation & Software Testing

Project: MediScript-E — Digital Healthcare Platform

Objectives

- Ensure correctness of MediScript-E through systematic testing
- Design comprehensive test cases for all major modules
- Create a Requirement Traceability Matrix (RTM)

Theory

- **Verification:** Building the system right — ensuring implementation matches design and specifications
- **Validation:** Building the right system — ensuring the system meets actual user needs

Testing improves confidence, not perfection. For MediScript-E, testing covers authentication, appointment management, prescription handling, medical vault, AI chatbot, search, and admin operations.

Task 1: Test Plan

1.1 Test Scope

The following modules are covered:

- Authentication (Registration, Login, Email Verification, OAuth, 2FA)
- Patient Module (Appointments, Prescriptions, Medicine Reminders, Medical Vault)
- Doctor Module (Appointment Management, Prescription Issuance, Archive/Edit/Delete)
- Admin Module (Dashboard, User Management, Appointment Overview)
- AI Chatbot Module (MediBot)
- Search Module (Global Search)
- Settings Module (Profile Update, Password Change, 2FA Toggle)

1.2 Test Types

Test Type	Description	Tool/Method
Unit Testing	Test individual API routes and functions	Manual / Jest
Integration Testing	Test interaction between modules	Manual API testing
Functional Testing	Test against functional requirements	Browser-based manual testing
Security Testing	Test authentication and authorization	Manual penetration testing
UI Testing	Test responsiveness and user interactions	Browser DevTools

1.3 Test Environment

Environment	Details
Local	<code>http://localhost:3000</code> with <code>.env</code> configuration
Production	<code>https://mediscript-e.vercel.app</code>
Database	Supabase PostgreSQL
Browser	Chrome, Firefox, Safari

1.4 Entry and Exit Criteria

Criteria	Details
Entry	All features implemented, build successful (<code>pnpm run build</code>)
Exit	All High priority test cases pass, no critical bugs

Task 2: Test Cases

Module 1: Authentication

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-01	Register with valid credentials (Patient)	Valid name, email, password, role=PATIENT, bloodGroup=O+	Account created, verification email sent	High	Pass
TC-02	Register with valid credentials (Doctor)	Valid name, email, password, role=DOCTOR, licenseNo, specialization	Account created, verification email sent	High	Pass
TC-03	Register with existing email	Already registered email	Error: "Email already registered"	High	Pass
TC-04	Register with invalid email format	<code>notanemail</code>	Validation error shown	High	Pass
TC-05	Register with short password	Password < 6 characters	Error shown	High	Pass
TC-06	Verify email with valid token	Valid token from email	<code>emailVerified: true</code> , redirect to login	High	Pass

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-07	Verify email with expired token	Token older than 24 hours	Error: "Verification link expired"	High	Pass
TC-08	Login with valid credentials	Verified email + correct password	JWT session created, redirect to /dashboard	High	Pass
TC-09	Login with wrong password	Valid email + wrong password	Error: "Invalid Email or Password"	High	Pass
TC-10	Login with unverified email	Unverified account credentials	Error: "Please verify your email"	High	Pass
TC-11	Login with Google OAuth	Valid Google account	Account created (if new), profile picture in session, redirect to /dashboard	Medium	Pass
TC-12	Login with GitHub OAuth	Valid GitHub account	Account created (if new), profile picture in session, redirect to /dashboard	Medium	Pass
TC-13	Profile picture displayed after OAuth login	Google/GitHub OAuth login	Profile picture shown in navbar and sidebar	Medium	Pass
TC-14	Enable 2FA from settings	Toggle 2FA on	<code>twoFactorEnabled: true</code> saved	High	Pass
TC-15	Login with 2FA enabled	Valid credentials + 2FA enabled	OTP sent to email, redirect to /verify-2fa	High	Pass
TC-16	Verify OTP with correct code	Valid 6-digit OTP	Session created, redirect to /dashboard	High	Pass
TC-17	Verify OTP with wrong code	Invalid OTP	Error: "Invalid OTP"	High	Pass
TC-18	Verify OTP after expiry	OTP older than 10 minutes	Error: "OTP has expired"	High	Pass
TC-19	Resend OTP	Click resend after 60 seconds	New OTP sent, countdown resets	Medium	Pass

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-20	Direct 2FA bypass attempt	POST signIn with <u>__2fa_verified__</u> without OTP	Error: "Invalid credentials"	High	Pass

Module 2: Patient — Appointments

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-21	Navigate to /appointments page	Logged in as patient	Appointments page loads with booking form and list	High	Pass
TC-22	Book appointment with valid data	Doctor, date, time, reason	Appointment created with status <u>PENDING</u>	High	Pass
TC-23	Book appointment without selecting doctor	No doctor selected	Validation error	High	Pass
TC-24	View all appointments with filter	Filter by PENDING	Only pending appointments shown	High	Pass
TC-25	Cancel pending appointment	Appointment with status <u>PENDING</u>	Status updated to <u>CANCELLED</u>	Medium	Pass

Module 3: Doctor — Appointments & Prescriptions

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-26	Navigate to /appointments page	Logged in as doctor	Doctor appointments page loads with filter tabs	High	Pass
TC-27	Default filter shows PENDING	Page load	PENDING tab selected by default	Medium	Pass
TC-28	Confirm pending appointment	Appointment with status <u>PENDING</u>	Status updated to <u>CONFIRMED</u>	High	Pass
TC-29	Cancel appointment with confirmation	Any active appointment	Confirmation dialog shown, then status updated to <u>CANCELLED</u>	High	Pass
TC-30	Complete confirmed appointment with confirmation	Appointment with status <u>CONFIRMED</u>	Confirmation dialog shown, then status updated to <u>COMPLETED</u>	High	Pass

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-31	Navigate to /prescriptions page	Logged in as doctor	Prescription form and list shown	High	Pass
TC-32	Issue prescription with valid data	Valid patient, diagnosis, medications	Prescription created, appointment auto-completed	High	Pass
TC-33	Issue prescription auto-completes appointment	Issue prescription for confirmed appointment	Related appointment status → COMPLETED	High	Pass
TC-34	Issue prescription with missing fields	Empty diagnosis or medications	Validation error	High	Pass
TC-35	Archive prescription	Active prescription	Prescription moves to Archived tab	Medium	Pass
TC-36	Unarchive prescription	Archived prescription	Prescription moves back to Active tab	Medium	Pass
TC-37	Edit prescription	Active prescription	Inline edit form shown, changes saved	Medium	Pass
TC-38	Delete prescription with confirmation	Any prescription	Confirmation dialog shown, then prescription deleted	Medium	Pass

Module 4: Patient — Prescriptions & Medical Vault

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-39	Navigate to /prescriptions page	Logged in as patient	All prescriptions displayed	High	Pass
TC-40	Download prescription as PDF	Click download on prescription	PDF generated and downloaded	High	Pass
TC-41	Navigate to /vault page	Logged in as patient	Medical vault page loads	Medium	Pass
TC-42	Upload medical document	Valid file	File uploaded to Supabase Storage, record saved	Medium	Pass
TC-43	Delete medical document	Existing document	Document deleted from storage and database	Low	Pass

Module 5: Medicine Reminders

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-44	Navigate to /reminders page	Logged in as patient	Reminders page loads	Medium	Pass
TC-45	Add medicine reminder with valid data	Medicine name, dosage, frequency, time, dates	Reminder created and displayed	Medium	Pass
TC-46	Mark medicine as taken	Active reminder	<code>taken: true,</code> <code>takenAt</code> timestamp saved	Medium	Pass
TC-47	Undo taken medicine	Taken reminder	<code>taken: false,</code> <code>takenAt</code> cleared	Medium	Pass
TC-48	Delete medicine reminder	Existing reminder	Reminder removed from list	Medium	Pass
TC-49	Automated email reminder	Cron job trigger	Email sent to patient for due reminders	Medium	Pass

Module 6: AI Chatbot (MediBot)

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-50	Open chatbot	Click floating chat button	Chat window opens with greeting	Medium	Pass
TC-51	Ask platform-related question	"What is MediScript-E?"	Accurate platform description in English	Medium	Pass
TC-52	Ask about features	"What can patients do?"	List of patient features in plain text	Medium	Pass
TC-53	Ask unrelated question	"What is the capital of France?"	Redirected to platform topics	Medium	Pass
TC-54	Ask for medical advice	"What medicine should I take for fever?"	Recommends consulting a real doctor	Medium	Pass
TC-55	Ask in Bengali	"MediScript-E কী?"	Response in English only	Medium	Pass
TC-56	Conversation history maintained	Multiple messages	Context maintained across messages	Low	Pass

Module 7: Global Search

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-57	Search as patient	Type doctor name	Matching appointments shown	Medium	Pass
TC-58	Search as doctor	Type patient name	Matching patients and appointments shown	Medium	Pass
TC-59	Search as admin	Type user email	Matching users shown	Medium	Pass
TC-60	Empty search	Clear search input	Dropdown closes	Low	Pass
TC-61	No results	Random string	"No results found." shown	Low	Pass
TC-62	Click search result	Click appointment result	Scrolls to appointments section	Low	Pass

Module 8: Admin

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-63	View admin dashboard	Logged in as admin	Real-time stats displayed (users, patients, doctors, appointments, prescriptions, contacts)	High	Pass
TC-64	Navigate to /users page	Logged in as admin	All users listed with profiles and roles	High	Pass
TC-65	Delete another user	Valid user ID (not self)	User deleted from database	High	Pass
TC-66	Delete own admin account	Admin's own user ID	Error: Cannot delete own account	High	Pass
TC-67	Navigate to /appointments page	Logged in as admin	All appointments with filter	Medium	Pass
TC-68	Navigate to /contacts page	Logged in as admin	All contact form submissions displayed	Low	Pass

Module 9: Settings

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-69	Update profile name	New valid name	Name updated in database and session	Medium	Pass
TC-70	Change password with correct current password	Valid current + new password	Password updated successfully	Medium	Pass
TC-71	Change password with wrong current password	Invalid current password	Error: "Current password is incorrect"	Medium	Pass
TC-72	Change password with mismatched new passwords	New $\neq$ confirm password	Error: "Passwords do not match"	Medium	Pass
TC-73	Toggle 2FA on	2FA currently disabled	<code>twoFactorEnabled: true</code> saved	Medium	Pass
TC-74	Toggle 2FA off	2FA currently enabled	<code>twoFactorEnabled: false</code> saved	Medium	Pass

Module 10: Security Testing

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-75	Access protected API without session	GET <code>/api/appointment</code> without auth	401 Unauthorized	High	Pass
TC-76	Access admin API as patient	GET <code>/api/admin/stats</code> as patient	403 Forbidden	High	Pass
TC-77	Access admin API as doctor	DELETE <code>/api/admin/users/[id]</code> as doctor	403 Forbidden	High	Pass
TC-78	SQL injection attempt	<code>' OR 1=1 --</code> in login email field	Prisma rejects, no data exposed	High	Pass
TC-79	Access another patient's data	Patient A accessing Patient B's appointments	Only own data returned	High	Pass
TC-80	Search cross-user data	Patient searching for another patient's records	Only own data returned	High	Pass
TC-81	Access patient-only page as doctor	Doctor navigating to <code>/reminders</code>	Redirect to <code>/dashboard</code>	High	Pass

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-82	Access admin-only page as patient	Patient navigating to /users	Redirect to /dashboard	High	Pass

Task 3: Requirement Traceability Matrix (RTM)

FR ID	Requirement	Test Case(s)	Status
FR-01	User Registration	TC-01, TC-02, TC-03, TC-04, TC-05	Pass
FR-02	Email Verification	TC-06, TC-07	Pass
FR-03	Credential Login	TC-08, TC-09, TC-10	Pass
FR-04	OAuth Login with Profile Picture	TC-11, TC-12, TC-13	Pass
FR-05	2FA via Email OTP	TC-14, TC-15, TC-16, TC-17, TC-18, TC-19, TC-20	Pass
FR-06	Book Appointment	TC-21, TC-22, TC-23	Pass
FR-07	Manage Appointments	TC-26, TC-27, TC-28, TC-29, TC-30	Pass
FR-08	Issue Prescription	TC-31, TC-32, TC-33, TC-34	Pass
FR-09	View/Download Prescription	TC-39, TC-40	Pass
FR-10	Medicine Reminders	TC-44, TC-45, TC-46, TC-47, TC-48	Pass
FR-11	Medical Vault Upload	TC-41, TC-42	Pass
FR-12	Delete Medical Report	TC-43	Pass
FR-13	Profile/Password Update	TC-69, TC-70, TC-71, TC-72	Pass
FR-14	2FA Toggle in Settings	TC-73, TC-74	Pass
FR-15	Admin Dashboard Stats	TC-63	Pass
FR-16	Admin User Management	TC-64, TC-65, TC-66	Pass
FR-17	Admin Appointment Monitor	TC-67	Pass
FR-18	Admin Contact Messages	TC-68	Pass
FR-20	Automated Reminders (Cron)	TC-49	Pass
FR-21	Prescription Archive	TC-35, TC-36	Pass
FR-22	Prescription Edit/Delete	TC-37, TC-38	Pass
FR-23	AI Chatbot (MediBot)	TC-50, TC-51, TC-52, TC-53, TC-54, TC-55, TC-56	Pass
FR-24	Global Search	TC-57, TC-58, TC-59, TC-60, TC-61, TC-62	Pass

FR ID	Requirement	Test Case(s)	Status
FR-25	OAuth Profile Pictures	TC-13	Pass
FR-26	Dedicated Feature Pages	TC-21, TC-26, TC-31, TC-41, TC-44, TC-64, TC-67, TC-68	Pass
NFR-08	RBAC on APIs	TC-75, TC-76, TC-77, TC-81, TC-82	Pass
NFR-13	Input Validation	TC-78	Pass
NFR-15	Role-based search	TC-79, TC-80	Pass

Key Findings / Learning Outcomes

- Designed **82 test cases** covering all 10 modules of MediScript-E
- New modules tested: AI Chatbot (TC-50 to TC-56), Global Search (TC-57 to TC-62), Security for new routes (TC-81, TC-82)
- Understood the difference between **verification** (correct implementation) and **validation** (correct system)
- Learned that AI chatbot requires specific test cases for content safety and language enforcement
- RTM ensures every functional requirement has at least one corresponding test case
- Recognized that dedicated routes require additional security tests to prevent unauthorized access
- Testing revealed that prescription auto-complete on issuance is a critical integration test